
Sean Webb

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EDUCATION

University of Colorado, Boulder - August 2022 - Present (Graduating May 2026)

B.S. Electrical & Computer Engineering | Engineering Physics | Minor: Quantum Engineering
Current GPA: 3.449

University of Colorado, Boulder - August 2025 - Present (Graduating May 2027)

M.S. Electrical Engineering (Embedded Systems Engineering Subplan)

SKILLS

Languages: C, C++, Python, Assembly (NIOS-II, RISC-V), HTML/CSS/JS, Verilog, System Verilog

Embedded Systems: Bare-metal programming, STM32 (F4 series), 8051, PCB Design (Altium Designer, KiCad), ARM Cortex-M, ADC/DAC interfacing, I2C/UART communication, Signal Integrity Optimization, Thread synchronization, Mutexes, Semaphores, RTOS task scheduling, HAL design and implementation, Vivado Design Suite

Systems: UNIX/Linux, OS Structure, Creating System Calls and Drivers

Tools: Git, GitHub, LaTeX, IBM Cloud AI, Oscilloscopes, Logic Analyzers

Quantum: Qiskit, Quantum Computer Architecture, Neutral Atom Trapping

PROJECTS

- Impedance Analyzer and Current Burn-In Tester for Atom Chip (Inflection Capstone)
 - Designed Raspberry Pi-based automated impedance measurement system achieving <1% error across 0.5Ω-3MΩ range
 - Implemented 4-wire (Kelvin) measurement technique for precision impedance matching
 - Developed high current burn-in testing framework with real-time thermal monitoring
 - Contributing to validation workflow for neutral atom trapping quantum hardware
- 8051 Multi-Function Embedded System
 - Developed interrupt-driven firmware on 8051 and STM32 microcontrollers
 - Implemented UART, GPIO, I2C, and timer drivers at register level
- “Golden” Arduino
 - Designed and fabricated a low-noise Arduino-compatible board from scratch and optimized signal integrity for precision measurements
 - Performed PCB layout in Altium
- 3RPS Parallel Manipulator
 - Implemented closed-loop PID control with real-time motor feedback
 - Processed touchscreen input to compute inverse kinematics and convert those kinematics into motor movement to stabilize a steel ball under dynamic disturbances

EXPERIENCE

Partnerships for Informal Science Education in the Community - *Student Volunteer*

JANUARY 2025 - MAY 2025

- Connected with local K-12 students to help engage them in inquiry-based science activities centered around physics

CU Boulder Bookstore, Boulder - *Student Accountant*

SEPTEMBER 2023 - JUNE 2024

- Managed billing reconciliation and financial record processing using Excel
- Processed and verified daily transaction reports and digitized records

Innovation Center, Longmont - *Development Event Organizer*

AUGUST 2021 - MAY 2022

- Team development using Python, LaTeX, and GitHub
- Event organization

Innovation Center, Longmont - *IBM Watson Development Team*

AUGUST 2020 - MAY 2022

- Designed and implemented AI Chatbots using IBM Watson for business and nonprofit clients, often communicating with the business owners directly

AWARDS

Quantum Forge Innovation Award (Capstone Award for Inflection Chip Testing Project)

RELEVANT COURSEWORK

Embedded/Computer Engineering:

- Embedded System Design
- Principles of Embedded Software
- Embedded Software Engineering
- Practical PCB Design and Fabrication
- Computer Organization
- Electromagnetic Fields and Waves
- Quantum Forge (Capstone with Inflection)
- Computer Science I & II (Data Structures)

Engineering Physics/Quantum Engineering:

- Quantum Mechanics I & II
- Foundations of Quantum Hardware
- Principles of Electricity and Magnetism I & II